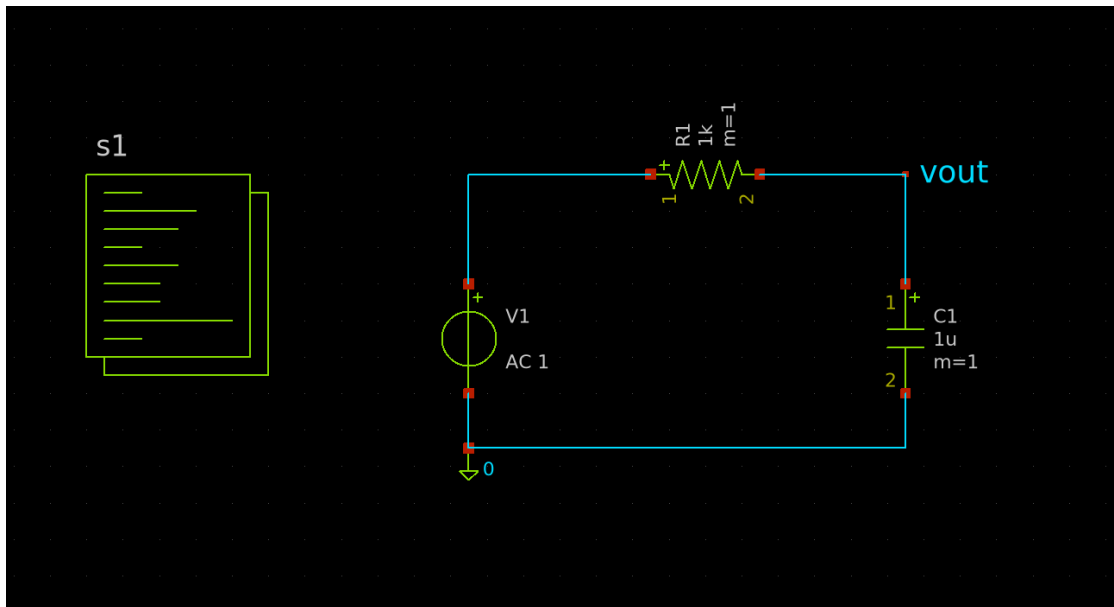


RC LOW PASS FILTER

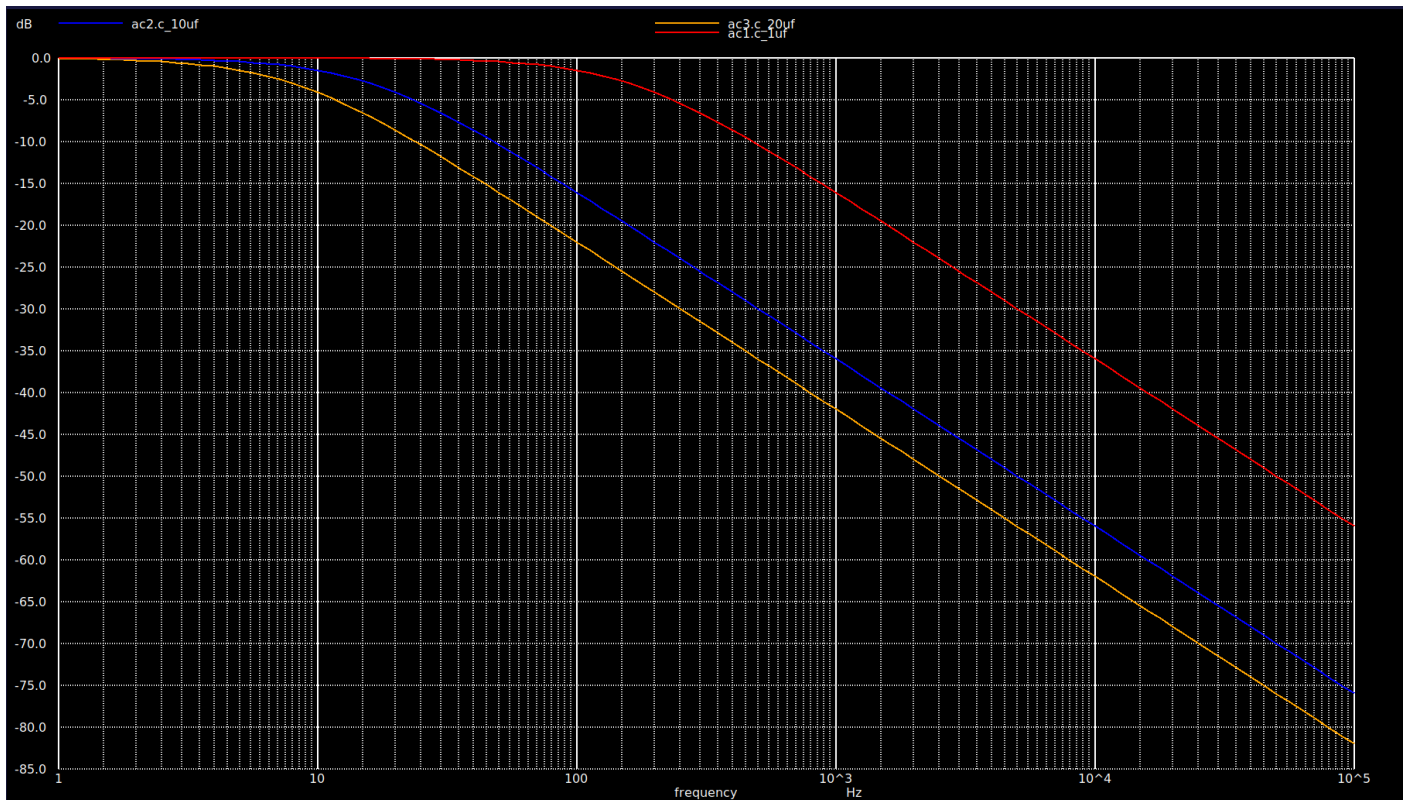
Github repo link:

https://github.com/Abhiram-Jayakumar-07/AnalogElectronicsLab_2025BEC0020/tree/main/RC%20Low%20Pass%20Filter

Schematic:



AC Analysis:



Results:

Capacitance (C)	Theoretical Bandwidth ($f_c = \frac{1}{2\pi RC}$)	Simulated Bandwidth
1 μF	159.15 Hz	158.784 Hz
10 μF	15.91 Hz	15.941 Hz
20 μF	7.96 Hz	8.064 Hz

Conclusion:

The AC frequency response analysis demonstrates that the cutoff frequency (f_{-3dB}) of an RC low-pass filter is inversely proportional to its capacitance. As the capacitance is increased from 1 μF to 10 μF to 20 μF , the filter's magnitude response curve shifts progressively lower on the frequency axis, shrinking the passband from 158.78 Hz down to 15.94 Hz and finally to 8.06Hz, respectively. Physically, a larger capacitance increases the time constant ($\tau = RC$), slowing down the circuit's voltage response and forcing attenuation to begin at much lower input frequencies. The simulated bandwidth values align almost identically with the theoretical predictions (159.15 Hz, 15.92 Hz, and 7.96 Hz), verifying the filter's performance across all three test cases.